Given

0.85	-1.2	0.34	2.1
-0.07	0.91	-1.88	0.12
1.55	0.03	-0.44	-2.31
-0.18	1.03	0.77	0.55

Tasks

1. Scale factor. Compute S using symmetric per-tensor quantization: S = max(|W|) / 127. Show the max value and the computed S

S = max(|W|) / 127

S = 2.31 / 127 = 0.01818898

2.Quantize. Quantize each element: W_q = round(W / S). Clamp to [−128, 127]. Write out the full 4×4 INT8 matrix.

46.73159243	-65.97401284	18.69263697	115.4545225
-3.848484082	50.03029307	-103.3592868	6.597401284
85.21643325	1.649350321	-24.19047137	-126.9999747
-9.896101925	56.62769435	42.3333249	30.23808922

W_q = round(W / S)

47	-66	19	115
-4	50	-103	7
85	2	-24	-127
-10	57	42	30

3. Dequantize. Compute W_deq = W_q × S. Write out the 4×4 FP32 dequantized matrix.

0.854882060	-1.200472680	0.345590620	2.091732700
-0.072755920	0.909449000	-1.873464940	0.127322860
1.546063300	0.036377960	-0.436535520	-2.310000460
-0.181889800	1.036771860	0.763937160	0.545669400

4. Error analysis. Compute the per-element absolute error |W − W_deq|. Identify the element with the largest error and compute the Mean Absolute Error (MAE) across all 16 elements.

-0.004882060	0.000472680	-0.005590620	0.008267300
0.002755920	0.000551000	-0.006535060	-0.007322860
0.003936700	-0.006377960	-0.003464480	0.000000460
0.001889800	-0.006771860	0.006062840	0.004330600

Largest Error	0.0082673
Sum	0.0692122
MAE	0.0043257625

5. Bad scale experiment. Use S_bad = 0.01 (too small). Repeat quantization and dequantization. Compute the MAE. Explain in one sentence what goes wrong when S is too small.

Original Matrix

0.85	-1.2	0.34	2.1
-0.07	0.91	-1.88	0.12
1.55	0.03	-0.44	-2.31
-0.18	1.03	0.77	0.55

Original Matrix * 100

85	-120	34	210
-7	91	-188	12
155	3	-44	-231
-18	103	77	55

Clamp to INT8 range [-128, 127]

210 → 127
-188 → -128
155 → 127
-231 → -128

85	-120	34	127
-7	91	-128	12
127	3	-44	-128
-18	103	77	55

Dequantize

0.85	-1.2	0.34	1.27
-0.07	0.91	-1.28	0.12
1.27	0.03	-0.44	-1.28
-0.18	1.03	0.77	0.55

Error						
0	0	0	0.83			
0	0	-0.6	0			
0.28	0	0	-1.03			
0	0	0	0			
Largest Error						
2.10 → 1.27 error = 0.83						
-1.88 → -1.28 error = 0.60						
1.55 → 1.27 error = 0.28						
-2.31 → -1.28 error = 1.03						
Sum	2.74					
MAE	0.17125					
When S is too small, large values exceed the INT8 range and get clipped to -128 or 127, causing large dequantization error.						